SURGIVISION™

AI-Assisted Haptic & Holographic Surgical Guidance System

“Feel the Touch. See Beyond Sight. Operate with Confidence.”

PROJECT REPORT — REVIEW 3

Final Review Submission

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1. Introduction

SURGIVISION™ is an AI-assisted haptic and holographic surgical guidance system designed to support surgical planning and spatial decision-making through 3D anatomical visualization, medical-image analysis, proximity-based safety feedback, and simulated haptic interaction. The project addresses the difficulty of mentally reconstructing three-dimensional anatomy from flat CT/PET image slices by creating an interactive 3D representation of the anatomy.

The project has progressed through three review stages. Review 1 established the research and interface foundation. Review 2 produced the Unity-based liver visualization, C# telemetry/proximity logic and a dashboard demonstration. Review 3 focuses on the 2D medical-image to 3D reconstruction workflow and the practical use of the reconstructed anatomy within the existing Unity prototype.

For the current demonstration, the system is intentionally organized into two modular components: a Web Dashboard for patient/scan information, AI analysis, risk analysis and reporting, and a Unity Prototype for the interactive 3D liver, virtual surgical tool, risk zones, proximity monitoring and simulated haptic feedback. Full real-time Unity-to-web integration is future work.

2. Progress Across Reviews

Milestone	Focus	Outcome
Review 1 — Foundation	Research & Design	Anatomical dataset study and Figma UI wireframe construction.
Review 2 — Core Engine	Unity 3D & Dashboard	Unity 6 integration, C# telemetry/proximity logic, liver visualization, and dashboard demonstration.
Review 3 — Current Phase	2D→3D & Modular Prototype	MONAI-based segmentation/reconstruction workflow, 3D mesh generation, Unity liver prototype, and separate web dashboard.

The main Review 3 advancement is the shift from using an externally sourced liver mesh to a workflow in which medical-image data can be processed, segmented and reconstructed into 3D geometry before being used by the Unity prototype.

3. Objectives of Review 3

• Develop an AI-assisted 2D-to-3D workflow for medical scan data using MONAI.

• Preprocess DICOM CT/PET slice series and assemble them into a consistent 3D volume.

• Apply confidence calibration before generating 3D surfaces.

• Generate polygonal anatomical meshes using Marching Cubes and export them for Unity.

• Use the reconstructed anatomy within the existing Unity liver surgical prototype.

• Maintain tool-to-target proximity and SAFE / CAUTION / DANGER safety feedback.

• Retain the Web Dashboard as a separate monitoring and planning module.

• Document limitations and define the next phase of real-time integration and hardware development.

4. Current System Approach

Because real-time integration between a web application and Unity introduces additional communication, deployment and synchronization complexity, the current prototype uses a modular architecture. Both modules follow the same SURGIVISION™ workflow but can be developed and demonstrated independently.

Module	Current Contents
Web Dashboard	Patient & Scan, AI Analysis, Risk Analysis, Surgical Planning, Live Monitoring and Reports.
Unity Prototype	3D liver, virtual surgical tool, target/risk zones, tool-to-target distance, safety state and simulated haptic feedback.
Common Workflow	Medical scan → AI analysis/reconstruction → 3D anatomy → surgical planning/safety monitoring.
Future Integration	Real-time exchange of scan, confidence, distance, alerts and telemetry between web and Unity.

Architecture: SURGIVISION™ → Web Dashboard + Unity Prototype → common data/workflow. Full live two-way integration is not claimed for the current review.

5. Literature Review

The literature review is limited to four selected papers that directly support the current technical direction.

No.	Research Paper	Main Contribution	Relevance
[1]	Lorensen & Cline (1987), “Marching Cubes: A High Resolution 3D Surface Construction Algorithm”	Introduced Marching Cubes for extracting 3D surfaces from volumetric data.	Supports conversion of reconstructed medical volumes into 3D meshes.
[2]	Cardoso et al. (2022), “MONAI: An Open-Source Framework for Deep Learning in Healthcare”	Introduces MONAI for deep-learning workflows on medical imaging.	Supports AI-based segmentation and preprocessing in the 2D→3D pipeline.
[3]	Azimi (2012), “Interactive 3D Registration and Trajectory Evaluation for Image-Guided Intervention”	Focuses on interactive 3D registration and trajectory evaluation for image-guided intervention.	Supports the project’s spatial tool-path and proximity-safety concept.
[4]	Egger et al. (2017), “Interactive 3D Visualization and Augmented Reality in Surgical Planning and Guidance”	Demonstrates interactive 3D/AR visualization for surgical planning and guidance.	Supports the interactive Unity visualization and future immersive guidance direction.

The selected literature provides the technical foundation for 3D surface reconstruction, medical-image AI, spatial safety concepts and interactive 3D visualization.

6. System Architecture and End-to-End Workflow

Step	Stage	Description
01	2D Data Input	DICOM CT/PET slice series are loaded and validated.
02	Preprocessing	3D volume assembly, isotropic resampling, intensity scaling and orientation normalization.
03	AI Segmentation	MONAI-based segmentation produces organ/target probability maps.
04	Confidence Calibration	Low-confidence predictions are filtered using a selected threshold.
05	3D Surface Extraction	Marching Cubes converts the calibrated volume into polygonal surfaces.
06	Mesh Export	Organ and target meshes are exported as .obj files.
07	Unity Prototype	3D anatomy is imported into the Unity surgical environment.
08	Safety Interaction	Virtual tool distance and safety states are evaluated in the Unity prototype.
09	Web Dashboard	Dashboard presents patient/scan, AI, risk, monitoring and reporting information separately.

The pipeline begins with 2D medical imaging data and ends in two demonstration outputs: reconstructed anatomy for the Unity prototype and structured information for the Web Dashboard.

7. 2D-to-3D Conversion Using MONAI

7.1 Preprocessing

The input consists of a stack of DICOM slices. The series is assembled into a 3D volume and resampled to isotropic voxel spacing. Intensity scaling and orientation normalization are applied before inference. For PET-oriented workflows, SUV scaling can be applied.

7.2 AI Segmentation

A MONAI-based segmentation workflow produces per-voxel probability maps for the organ and target region. Sliding-window inference can be used for larger volumes.

7.3 Confidence Calibration

A confidence threshold T in the range 0.6–0.8 is used as the current calibration range. Low-confidence voxels are filtered before surface extraction. Project targets are Dice > 0.85, IoU > 0.75 and false positives < 5%.

Metric	Target
Dice Score	> 0.85
IoU	> 0.75
False Positives	< 5%
Confidence Threshold T	0.6 – 0.8

7.4 Surface Extraction and Export

After calibration, the volume is converted to a triangle mesh using Marching Cubes. The mesh can be smoothed/decimated for real-time rendering and exported as .obj files for Unity.

8. Unity 3D Prototype — Current Liver Implementation

The current Unity demonstration is liver-based. The prototype contains a 3D liver anatomy, a virtual surgical tool, multiple safety/risk zones, a selected target, distance monitoring, a safety-state indicator and simulated haptic feedback.

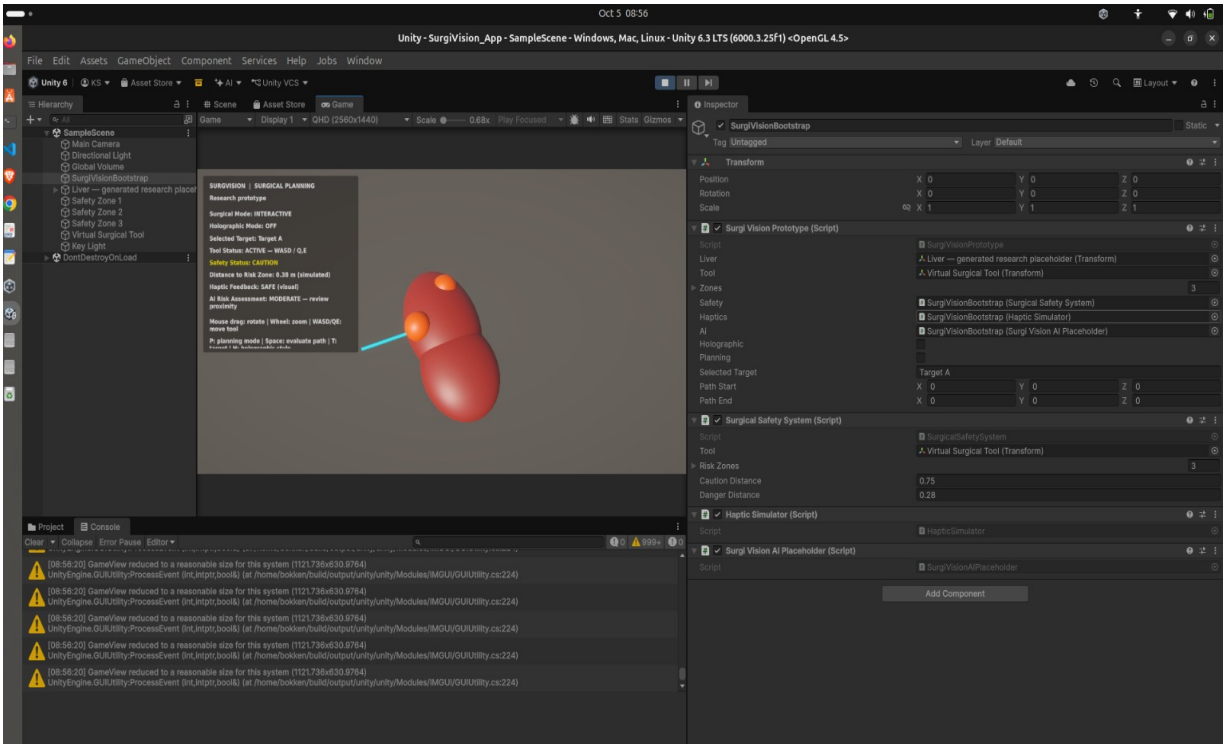


Figure 1 — Current SURGIVISION™ Unity liver surgical prototype.

8.1 Unity Components

• 3D Liver Anatomy — interactive anatomical representation used for the current surgical scenario.

• Virtual Surgical Tool — simulated instrument that can be moved within the scene.

• Risk/Safety Zones — predefined regions used for proximity-based safety evaluation.

• Target Selection — identifies the current surgical target.

• Distance Monitoring — computes the spatial relationship between tool and target/risk geometry.

• Safety State — maps proximity to SAFE, CAUTION or DANGER.

• Haptic Simulation — software-based feedback near the safety boundary.

• AI Placeholder/Interface — reserved for deeper AI-driven safety and planning integration.

8.2 Proximity and Safety Logic

The C# interaction logic computes tool-to-target/target-geometry distance and maps it to a safety state. When the distance enters a defined boundary, simulated feedback and logging can be triggered.

```
dist = Vector3.Distance(tool.pos, target.pos);
if dist < threshold → TriggerHaptic(dist) → Log("Safety Boundary Breach")
```

9. Web Dashboard — Current Modular Component

The Web Dashboard is treated as a separate application rather than being embedded inside Unity. Its purpose is to provide the surrounding workflow for scan management, AI analysis, risk information, monitoring and reports.

Page/Module	Purpose
Dashboard	Case overview, scan type, AI status, reconstruction status, risk status and recent alerts.
Patient & Scan	Demo patient/case information and scan input/processing controls.
AI Analysis	Preprocessing → MONAI segmentation → confidence calibration → 3D reconstruction.
Surgical Planning	Target, risk zones, safety state, tool distance, AI assessment and planned path information.
Live Monitoring	Tool status, target, distance, safety state, AI risk and event timeline.
Reports	Previous cases, safety events, confidence and planning/session summaries.

For the current review, the dashboard is not claimed to have a live two-way connection with Unity. The two components are aligned through the same data model and workflow. Real-time communication is future work.

10. Technical Modules

Module	Core Technology	Current Function
Web Dashboard	React/Vite-based monitoring and planning interface	Patient/Scan, AI Analysis, Risk Analysis, Reports
Medical Image Pipeline	MONAI + preprocessing + confidence calibration	2D scan → segmented volume
3D Reconstruction	Marching Cubes + mesh export	3D organ/target geometry
Unity Prototype	Unity 6 + C# interaction logic	3D liver, surgical tool, risk zones, safety state
Haptic Simulation	Software simulation	Visual/simulated feedback near safety boundary
Telemetry / Logging	Prototype event and state logging	Distance, safety state and alerts

11. Key Challenges and Implemented Solutions

1. Challenge: 2D scan data is not directly usable as a 3D surgical model.

Solution: Assemble the slice series into a 3D volume, normalize spacing/orientation, apply preprocessing, segment the target, and extract a surface mesh.

2. Challenge: Segmentation errors can create incorrect anatomy.

Solution: Use MONAI-based segmentation with probability maps and confidence thresholding before mesh generation.

3. Challenge: A 3D model alone does not provide surgical safety feedback.

Solution: Calculate tool-to-target proximity and map it to SAFE, CAUTION and DANGER states.

4. Challenge: Full real-time dashboard–Unity integration is technically complex for the current review.

Solution: Use a modular architecture: develop and demonstrate the web dashboard and Unity prototype separately, with a common workflow/data model. Full live integration remains future work.

5. Challenge: True haptic and mixed-reality hardware is not available in the current prototype.

Solution: Use software-based haptic simulation and holographic-style visualization on a standard display, while keeping physical hardware as future work.

12. Results and Demonstration

The current Review 3 demonstration follows the path from medical-image processing to the two modular outputs.

• Medical-image slices are processed into a 3D volume.

• MONAI-based segmentation and confidence filtering are applied.

• The calibrated volume is converted into 3D surface geometry and exported.

• The 3D geometry is used within the Unity surgical prototype.

• The Unity scene presents the liver, virtual tool, risk zones and target.

• Tool-to-target proximity is evaluated and mapped to a safety state.

• Simulated haptic/visual feedback is triggered near safety boundaries.

• The Web Dashboard separately presents the patient, AI, risk and reporting workflow.

12.1 Review 2 vs Review 3

Aspect	Review 2	Review 3
3D anatomy source	Open-access liver mesh	Project reconstruction workflow + 3D mesh generation
AI component	Planned	MONAI-based segmentation/calibration workflow
Unity model	Holographic-style liver + proximity script	Liver prototype with tool, risk zones, distance and safety feedback
Dashboard	Presented	Retained as a separate web module
Integration	Conceptual ecosystem	Modular approach; full real-time Unity↔web integration is future work

13. Limitations and Future Work

SURGIVISION™ remains an academic research prototype. It is not a medical device, has not been clinically validated, and must not be used for diagnosis or treatment decisions. The current 'holographic' view is a holographic-style 3D visualization on a standard display, and haptic feedback is simulated in software.

• Validate segmentation on larger, annotated datasets and report Dice, IoU and Hausdorff metrics.

• Complete robust real-time communication between the Web Dashboard and Unity prototype.

• Connect physical haptic hardware and a head-mounted display for true mixed-reality interaction.

• Integrate live tracking and future hospital/PACS data ingestion.

• Expand beyond the current liver scenario to brain, cardiac and spine anatomies.

• Strengthen logging, session replay and quantitative safety evaluation.

14. Conclusion

In Review 3, SURGIVISION™ progressed from a Unity visualization concept toward a more complete medical-image-to-3D prototype workflow. The project incorporates MONAI-based medical-image segmentation, confidence calibration and Marching Cubes surface reconstruction as the technical path from 2D scan data to 3D geometry. The reconstructed anatomy is intended for use within the existing Unity liver surgical environment, where tool proximity, risk zones, safety states and simulated haptic feedback can be demonstrated.

The current implementation deliberately follows a modular approach: the Web Dashboard and Unity Surgical Prototype are maintained as separate components because full real-time integration is beyond the current prototype scope. This allows both the monitoring workflow and the interactive 3D surgical environment to be demonstrated while leaving real-time communication, physical haptics and clinical validation as future development.

15. References — IEEE Style

[1] W. E. Lorensen and H. E. Cline, “Marching Cubes: A High Resolution 3D Surface Construction Algorithm,” ACM SIGGRAPH Computer Graphics, vol. 21, no. 4, pp. 163–169, 1987.

[2] M. J. Cardoso et al., “MONAI: An Open-Source Framework for Deep Learning in Healthcare,” arXiv:2211.02701, 2022.

[3] E. Azimi, “Interactive 3D Registration and Trajectory Evaluation for Image-Guided Intervention,” IEEE Transactions on Biomedical Engineering, vol. 59, no. 12, pp. 3485–3495, 2012.

[4] J. Egger et al., “Interactive 3D Visualization and Augmented Reality in Surgical Planning and Guidance,” Computers in Biology and Medicine, vol. 89, pp. 122–131, 2017.